Haar Sampling and CNN Robustness: How Dense Should Rotation Augmentation Be?

Phillip Korolev
pkoro029@uottawa.ca

May, 2026

Abstract

Standard convolutional neural networks (CNNs) are not rotation invariant. The conventional remedy is data augmentation: at each training step, sample a random rotation and update on the transformed image. In practice, only *one* transformation is drawn per image per gradient step. This paper investigates what happens when we raise that count to $m > 1$ independent samples drawn from the Haar measure of a rotation group, and examines how the resulting sampling density m trades off accuracy and robustness gains against a proportionally growing compute cost. Framing the empirical orbit average as a Monte Carlo estimator of the group-invariant Haar projection, we establish that improvements follow an $O(1/\sqrt{m})$ rate: the first few doublings of m yield substantial gains, but returns diminish rapidly while cost scales linearly in m. For finite groups C_k we characterize a combinatorial saturation threshold beyond which all group elements are reliably covered and additional samples become redundant; for the continuous group $\mathrm{SO}(2)$ the saturation is softer, governed by orbit-gradient variance reduction. We evaluate these predictions on MNIST across $m \in \{1, 2, 4, 8, 16\}$ under both $\mathrm{SO}(2)$ and discrete C_k augmentation, measuring test accuracy, rotation robustness across all 360°, and per-epoch training cost. The results confirm the predicted diminishing-returns profile and identify $m \in \{4, 8\}$ as the practical operating point: beyond $m = 8$, additional samples provide less than 0.5% robustness gain at twice the compute cost.

1 Introduction

A central challenge in visual recognition is building models that respond correctly to a pattern regardless of how it is oriented on the input plane. This is a limitation of convolutional neural networks. The convolution operation grants translation equivariance: a shift in the input produces an identical shift in every feature map, a consequence of weight sharing across spatial positions. However, rotating an image by even a modest angle changes the activations throughout the network, and the final classification can flip entirely on inputs the model has not seen in that orientation. For tasks where the orientation of an object carries no label information like handwritten digit recognition, medical cell imaging, and aerial scene classification, this is a genuine failure mode rather than an acceptable limitation.

Two broad strategies address this gap. The first is *architectural equivariance*: redesigning the network so that the group of rotations is built directly into its layers. Group equivariant CNNs (G-CNNs) [5] and steerable CNNs [18, 6] achieve this by replacing standard convolution with group convolution, which commutes

exactly with the chosen symmetry group. The cost is heavy re-engineering and oftentimes, increased memory consumption. The second strategy is *data augmentation*: training a standard CNN on randomly transformed versions of each input and relying on the optimizer to internalize the implied invariance. Augmentation is a practical choice as it requires no architectural changes to the model. What augmentation-based approaches have not systematically examined, however, is the *density* of the sampling. Standard practice draws one random rotation per training image per gradient step (sampling density $m = 1$). Increasing m means presenting the model with m independently rotated copies of every image within each training step, explicitly averaging their gradients. With increased sampling during training, our model simultaneously sees multiple viewpoints of the same pattern, pushing the gradient update toward the orbit-averaged loss rather than a noisy single-sample estimate.

Each additional sample multiplies the per-step cost by a constant factor, so a large m is expensive. The question is whether the accuracy gains justify the cost, and how rapidly those gains diminish as m grows. We argue that this question has a precise mathematical structure. Using the empirical orbit average as a Monte Carlo estimator of the group-invariant Haar projection, the approximation error decays as $O(1/\sqrt{m})$ — an initially large performance boost that eventually becomes unnoticeable as $m \to \infty$. What it leaves open are the group-specific constants: a finite group C_k saturates in a different way than the continuous group $\mathrm{SO}(2)$, and the relevant thresholds are what practitioners need to know.

1.1 Problem Statement and Contributions

Let G be a compact rotation group, $\mathcal{X}$ an image space, and let $f_\theta : \mathcal{X} \to \mathbb{R}^C$ be a CNN producing class logits. For each training image x we draw m i.i.d. samples $g_1, \ldots, g_m$ from the Haar probability measure μ_G and form the per-image training objective by averaging the cross-entropy loss across the transformed copies $g_1x, ..., g_mx$. We study the following:

Q1. Accuracy scaling. How does test accuracy and rotation robustness scale with m? Does the scaling match the $O(1/\sqrt{m})$ rate predicted by Monte Carlo theory?

Q2. Group dependence. How does the answer to **Q1** depend on whether G is a finite group C_k or the continuous group $\mathrm{SO}(2)$?

Q3. Saturation threshold. For each group, what is the smallest m beyond which further gains are negligible relative to the additional compute?

Q4. Optimization effect. Does increasing m reduce gradient variance during training in the theoretically predicted manner, and is this visible in training dynamics?

In effort to answer these questions, we make the following four contributions.

- **Hoeffding concentration bound for** $\mathrm{SO}(2)$ (Section 2.4). We prove that with probability at least $1 - \delta$:

$$m \geq \frac{(b-a)^2}{2\varepsilon^2} \ln \frac{2C}{\delta}$$

  samples allow the empirical Haar projection to deviate from the true projection by at most ε in sup-norm. We translate this into a loss bound via the Lipschitz property of cross-entropy.

- **Orbit-gradient variance reduction** (Section 2.5). We prove that the variance of the Monte Carlo gradient estimator decays exactly as σ^2/m, providing an optimization rationale for the empirical model accuracy gains associated with larger m.

- **Empirical evaluation on MNIST** (Section 4). We evaluate $m \in \{1, 2, 4, 8, 16\}$ under $\mathrm{SO}(2)$ and discrete C_k augmentation, measuring test accuracy, rotation robustness across all 360 degrees of rotation, orbit-gradient variance across training epochs, and expected calibration error. We identify $m \in \{4, 8\}$ as the practical operating range: $m = 4$ maximises accuracy per unit of compute, while $m = 8$ recovers an additional 0.4 pp at twice the cost. Beyond $m = 8$, gains reverse.

1.2 Related Work

Group-equivariant networks. Cohen and Welling [5] introduced G-CNNs, replacing standard spatial convolution with group convolution to achieve exact equivariance. Weiler and Cesa [18] generalized this to the full $E(2)$-equivariant family through steerable filters, covering arbitrary combinations of rotation, reflection, and continuous symmetries. Kondor and Trivedi [11] showed that equivariance to a compact group is *necessary and sufficient* for a linear map between feature spaces to commute with group actions, grounding these architectures in a general framework. Our work takes the complementary route: rather than hard wiring invariance architecturally, we induce it through augmentation and study how many samples that requires.

Sample complexity of invariant learning. Bietti et al. [1] studied sample complexity when the ground truth is invariant to a compact group, proving tighter bounds for equivariant models relative to generic ones. Elesedy and Zaidi [7] proved strict generalization benefits for equivariant models under a symmetry assumption on the data distribution. Although the results established that invariance helps, our work asks *how much* of it to include per step.

Augmentation as invariance induction. Chen et al. [4] developed a group-theoretic framework for data augmentation in which orbit-averaging the loss is equivalent to projecting the model onto the space of invariant functions, providing the formal foundation our analysis builds on. Lyle et al. [13] showed, empirically and theoretically, that invariance to augmentation transformations improves generalization, while Botev et al. [2] proposed explicit regularization toward invariance as a complement to augmentation. Shanmugam et al. [16] studied aggregation strategies for test-time augmentation and found that the number of augmentation samples at inference follows similar diminishing-returns dynamics to those we study at training time. None of these works systematically varies the per-step sampling density m or characterizes the resulting cost–accuracy tradeoff, which is the gap our paper addresses.

2 Haar Sampling and Theoretical Density Bounds

We develop the mathematical framework governing Haar sampling and derive the density bounds that predict the empirical behavior studied in Section 4. Throughout the section, we work with compact rotation groups G acting on the image space $\mathcal{X} \subset \mathbb{R}^{H \times W}$.

2.1 The Haar Measure and Projection

Definition 2.1 (Haar probability measure)**.** The *Haar probability measure* μ_G on a compact group G is the unique Borel probability measure satisfying *left-invariance*: $\mu_G(hA) = \mu_G(A)$ for every $h \in G$ and every measurable $A \subseteq G$.

The existence and uniqueness of μ_G is a classical theorem due to Haar (1933) and Weil [8]. For $G = C_k$, μ_G is the uniform distribution on the k group elements, assigning mass $1/k$ to each. For the continuous rotation group $G = \mathrm{SO}(2)$, μ_G is the uniform distribution on $[0, 2\pi)$, with density $d\mu_G(\theta) = d\theta/(2\pi)$. Left-invariance is the key property we exploit: it ensures that a sample $g \sim \mu_G$ and a shifted sample $hg \sim \mu_G$ have the same distribution, which is what makes the Haar projection well-defined.

Let $f_\theta : \mathcal{X} \to \mathbb{R}^C$ be a CNN that takes as input image $x \in \mathcal{X}$ and returns class logits. We can define its

group-invariant counterpart by averaging predictions over the entire orbit – exactly the *Haar projection*.

Definition 2.2 (Haar projection)**.** The Haar projection (integral) of f_θ is

$$f^*(x) \;:=\; \int_G f_\theta(g \cdot x)\, d\mu_G(g).$$

Analytically, f^* is the orthogonal projection of f_θ onto the closed subspace of G-invariant functions in $L^2(\mathcal{X})$ [17, 14]. Furthermore, f^* is the closest invariant function to f_θ in L^2 distance, and we can show that the projection is linear, G-invariant, and idempotent.

Proof. Linearity follows from linearity of the integral. To show invariance with respect to G, take any $h \in G$. Then,

$$f^*(h \cdot x) = \int_G f_\theta(g \cdot (h \cdot x))\, d\mu_G(g) = \int_G f_\theta((gh) \cdot x)\, d\mu_G(g).$$

Substituting $g' = gh$ and using left-invariance of μ_G gives $f^*(h \cdot x) = f^*(x)$. Idempotency follows immediately: since f^* is already G-invariant, integrating over G recovers it, so $(f^*)^*(x) = \int_G f^*(g \cdot x)\, d\mu_G(g) = \int_G f^*(x)\, d\mu_G(g) = f^*(x)$. □

The Haar projection is crucial; a model that perfectly approximates f^* on all inputs would give rotation-invariant predictions by definition. Training with Haar sampling is a Monte Carlo procedure that pushes f_θ toward f^*. We examine how accurately this push lands as a function of m.

2.2 The Monte Carlo Estimator and Its Variance

Throughout computational experiments and in real-world applications, we must approximate the Haar projection using Monte Carlo averaging. Given m i.i.d. samples $g_1, ..., g_m \sim \mu_G$, the *empirical* projection is

$$f_m^*(x) \;:=\; \frac{1}{m} \sum_{i=1}^{m} f_\theta(g_i \cdot x).$$

This is an unbiased estimator with variance $\sigma_f^2(x)/m$, where $\sigma_f^2(x)$ is the orbit variance of f_θ at x, defined by

$$\sigma_f^2(x) := \int_G \|f_\theta(g \cdot x) - f^*(x)\|^2 \, d\mu_G(g)$$

Applying the strong law of large numbers to the i.i.d. sequence $f_\theta(g_i \cdot x)$ tells us that $f_m^* \to f^*$ almost surely as $m \to \infty$. We can quantify the distance $\|f_m^* - f^*\|$ in $L^2(\mathcal{X})$ to achieve a theoretical rate of this convergence. Applying Fubini's Theorem:

$$\mathbb{E}_x\Big[\|f_m^*(x) - f^*(x)\|^2\Big] = \mathbb{E}_x[\mathrm{Var}[f^*(x)]] = \frac{1}{m}\,\mathbb{E}_x\big[\sigma_f^2(x)\big] = O(1/m).$$

And thus:

$$\|f_m^* - f^*\|_{L^2} = O(1/\sqrt{m}).$$

The Monte Carlo estimator and its variance is the theoretical engine behind the empirical trends we observe: every doubling of m reduces the estimator's standard deviation by a factor of $1/\sqrt{2}$, predicting sharply diminishing returns in exactly the form we see throughout experiments. It also immediately raises the question: how does m interact with the structure of G? For a finite group C_k there is a combinatorial saturation phenomenon (once the model has seen every group element, adding samples becomes redundant in a precise sense) that does not exist for a continuous group like $\mathrm{SO}(2)$. The next two subsections derive bounds capturing these two regimes.

2.3 Finite Groups: A Coupon-Collector Bound

When we work with the group $G = C_k$, the Haar measure is the uniform distribution on k elements. The samples $g_1, \ldots, g_m$ are drawn independently and uniformly from $\{g_0, \ldots, g_{k-1}\}$, possibly with repetition. To ensure our CNN model trains on enough rotations of a particular image, we are curious to know how large must m be before every group element has been seen at least once with high probability? This is the *coupon collector* problem [15], and it gives a combinatorial threshold for m.

Theorem 2.3 (Sample size for full orbit coverage)**.** *Let $\delta \in (0, 1)$. If*

$$m \;\geq\; k\big(\ln k + \ln \tfrac{1}{\delta}\big)\,,$$

then with probability at least $1 - \delta$ every element of C_k appears at least once among $g_1, \ldots, g_m$.

Proof. Fix any element $g_j \in C_k$. The probability that g_j is *not* drawn in a single trial is $1 - 1/k$, so the probability that it is missed in all m independent trials is $(1 - 1/k)^m$. Applying the union bound over all k group elements, we get the probability that there exists some group element $g_j \in C_k$ which was left undrawn:

$$\mathbb{P}\left[\exists g_j \notin g_1, ..., g_m\right] \;\leq\; k\left(1 - \frac{1}{k}\right)^m \;\leq\; k\, e^{-m/k},$$

where the second inequality uses $1 - t \leq e^{-t}$ for $t \geq 0$. Setting $ke^{-m/k} \leq \delta$ and solving for m:

$$e^{-m/k} \leq \frac{\delta}{k} \;\;\Longrightarrow\;\; m \geq k \ln \frac{k}{\delta} = k\big(\ln k + \ln \tfrac{1}{\delta}\big)\,. \qquad \square$$

In the complementary direction, the expected number of unique elements in m draws from a finite collection of transformations $\{g_1, ..., g_k\}$ is

$$\mathbb{E}[\#\text{ unique}] = k\Big(1 - \big(\tfrac{k-1}{k}\big)^m\Big)\,.$$

Theorem 2.3 gives the m needed to *see* every group element. Full coverage is a sufficient condition for the orbit average to be unbiased with respect to every element, but as we showed, the empirical Haar projection f_m^* is unbiased for *all* $m \geq 1$. The coupon-collector threshold marks a qualitative change: below it, some rotations are systematically absent from any given training step; above it, all rotations are represented. This structural difference is what drives the empirical saturation we observe near $m \approx k$ when working with transformations sampled from finite groups.

2.4 Continuous Groups: A Hoeffding Concentration Bound

For $G = \mathrm{SO}(2)$ there is no notion of full coverage, so we cannot use the same framing as before. Instead, we bound the deviation of the empirical projection f_m^* from the true projection f^* using a concentration inequality. Let us assume that there exist constants $a < b$ such that every coordinate of $f_\theta(x)$ takes values in $[a, b]$ for all $x \in \mathcal{X}$. This holds whenever the input is bounded and the network weights have finite ℓ_∞ norm, or is enforced by logit clipping; it is standard in the generalization literature for bounded losses [3].

Theorem 2.4 (Concentration of the empirical Haar projection)**.** *Under the assumptions above, for every $x \in \mathcal{X}$, every class $c \in \{1, \dots, C\}$, and every $\varepsilon, \delta > 0$,*

$$\mathbb{P}[\,|[f_m^*(x)]_c - [f^*(x)]_c| > \varepsilon\,] \;\le\; 2\exp\!\left(-\frac{2m\varepsilon^2}{(b-a)^2}\right).$$

Equivalently, a coordinate-wise deviation of at most ε holds with probability at least $1-\delta$ provided

$$m \;\ge\; \frac{(b-a)^2}{2\varepsilon^2}\,\ln\frac{2}{\delta}. \tag{1}$$

Applying the union bound over all C coordinates, the sup-norm deviation $\|f_m^*(x) - f^*(x)\|_\infty \le \varepsilon$ holds with probability $\ge 1-\delta$ provided the same value of m in (1).

Proof. We fix x and class index c. The random variables $Z_i := [f_\theta(g_i \cdot x)]_c \in [a, b]$ are i.i.d. by the i.i.d. assumption on the g_i, with common mean $[f^*(x)]_c = \mathbb{E}[Z_i]$. Hoeffding's inequality [10] states that for i.i.d. random variables bounded in $[a, b]$,

$$\mathbb{P}\left[\left|\frac{1}{m}\sum_{i=1}^{m} Z_i - \mathbb{E}[Z_i]\right| > \varepsilon\right] \;\le\; 2\exp\!\left(-\frac{2m\varepsilon^2}{(b-a)^2}\right).$$

Since $[f_m^*(x)]_c = \frac{1}{m}\sum_i Z_i$, this is precisely the first displayed inequality. Solving for m in the right hand side gives (1). Applying the union bound over the C coordinates (each satisfying the per-coordinate bound) and setting the total failure probability to δ requires replacing δ with δ/C per coordinate, which yields the result. □

Corollary 2.5 (Loss concentration)**.** *If the cross-entropy loss $\ell(z, y) = -\log \mathrm{softmax}(z)_y$ is L-Lipschitz in z on $[a, b]^C$ (which holds with $L \le 1$ for softmax cross-entropy [9]), then under the conditions of Theorem 2.4,*

$$\mathbb{P}[\,|\ell(f_m^*(x), y) - \ell(f^*(x), y)| > L\varepsilon\,] \;\le\; 2C\exp\!\left(-\frac{2m\varepsilon^2}{(b-a)^2}\right).$$

Example 2.6 (Numerical estimate for $\mathrm{SO}(2)$ on MNIST)**.** Suppose logits are clipped to $[a, b] = [-5, 5]$, $C = 10$ classes, target deviation $\varepsilon = 0.1$, and confidence $1 - \delta = 0.95$. Then (1) gives $m \ge 5000 \cdot \ln 400 \approx 29{,}900$. This is far larger than the empirically effective range $m \in \{4, 8, 16\}$ verified in Section 4—an expected consequence of distribution-free Hoeffding bounds, which treat each Z_i as adversarially distributed across all of $[a, b]$ rather than exploiting the true orbit variance $\sigma_f^2(x)$. The bound's contribution is qualitative: it correctly predicts that m must grow quadratically to halve the approximation error, and it guarantees that the empirical Haar projection converges to the true projection, establishing that Haar sampling is a principled approximation procedure regardless of the group.

2.5 Orbit-Gradient Variance Reduction

Beyond approximation quality, Haar sampling has a direct effect on the optimization process. Each gradient step on the orbit-averaged loss is a Monte Carlo estimate of the true orbit gradient, and its variance—which governs the noisiness of SGD—decays inversely with m.

The per-image training objective under m-sample Haar averaging is

$$\hat{\mathcal{L}}^{(m)}(\theta;\, x, y) \;:=\; \frac{1}{m}\sum_{i=1}^{m} \ell(f_\theta(g_i \cdot x),\, y),$$

a Monte Carlo estimate of the orbit loss $\mathcal{L}_{\mathrm{orbit}}(\theta;\, x, y)$ in the continuous sense.

Theorem 2.7 (Orbit-gradient variance scales as $1/m$)**.** *Let* $\Sigma_g(\theta, x, y) := \mathrm{Var}_{g\sim\mu_G}\{\nabla_\theta \ell(f_\theta(g \cdot x), y)\}$ *denote the component-wise orbit variance of the per-sample gradient. Then*

$$\mathrm{Var}\Big\{\nabla_\theta \hat{\mathcal{L}}^{(m)}(\theta;\, x, y)\Big\} \;=\; \frac{1}{m}\Sigma_g(\theta, x, y).$$

Proof. By the i.i.d. property of the samples $g_1, \ldots, g_m$, the gradient of the estimator is a simple average of m independent, identically distributed random vectors:

$$\nabla_\theta \hat{\mathcal{L}}^{(m)} = \frac{1}{m}\sum_{i=1}^{m} \nabla_\theta \ell(f_\theta(g_i \cdot x), y).$$

For i.i.d. random vectors with component-wise variance Σ_g, the variance of the average is Σ_g/m:

$$\begin{aligned}
\mathrm{Var}\left\{\frac{1}{m}\sum_{i=1}^{m} \nabla_\theta \ell(f_\theta(g_i \cdot x), y)\right\} &= \frac{1}{m^2}\sum_{i=1}^{m} \mathrm{Var}\{\nabla_\theta \ell(f_\theta(g_i \cdot x), y)\} \\
&= \frac{1}{m^2} \cdot m \cdot \Sigma_g(\theta, x, y) \\
&= \frac{1}{m}\Sigma_g(\theta, x, y).
\end{aligned}$$

□

Remark 2.8. The standard $m = 1$ gradient has orbit variance Σ_g, which cannot be reduced without averaging. Theorem 2.7 says that using m samples per image acts as m-fold variance reduction on the orbit component of gradient noise—the component driven by the random choice of transformation. The variance from *mini-batch sampling* across images is unaffected. Concretely: doubling m halves the orbit-gradient noise; quadrupling m halves it again. This is the optimization rationale for the smoother training curves and lower epoch-to-epoch accuracy jitter we observe for large m in Section 4.

2.6 Summary of Theoretical Predictions

Table 1: Theoretical scaling laws and their empirical predictions.

Quantity	**Scaling in** m	**Empirical prediction**
$\|f_m^* - f^*\|_{L^2}$ (std. dev.)	$O(1/\sqrt{m})$	Robustness improvement slows with each doubling of m
$\mathrm{Var}(f_m^*)$	$\sigma_f^2(x)/m$	Prediction variance halves when m doubles
Expected unique coverage of C_k	$k(1-(1-1/k)^m)$	Performance saturates near $m \approx k \ln k$
Orbit-gradient variance	Σ_g/m	Training curves smooth out; less epoch-to-epoch jitter

A few qualitative predictions follow directly. First, the improvement from $m = 1$ to $m = 2$ should be the single largest jump–halving the approximation variance–with each subsequent doubling producing a smaller gain. Second, for C_k augmentation, performance should plateau near $m \approx k \ln k$, beyond which the coupon-collector threshold is reliably exceeded and additional samples are redundant. Lastly, orbit-gradient variance should visibly decrease across training epochs as m increases, which is directly measurable and provides an independent validation of the theory. We verify all three in Section 4.

3 Architecture and Experimental Design

Throughout experimentation, all conditions share a fixed CNN architecture, optimizer, and data split; only the Haar sampling regime varies. We train on the MNIST dataset [12] to classify handwritten digits, varying the sampling density m during training and evaluating on both normally-oriented and randomly rotated test images to measure rotation robustness. A *baseline* model trained without any rotation augmentation serves as the lower bound reference.

3.1 Setup

We use 8,000 training and 2,000 test images from MNIST [12] (28×28 greyscale, 10 classes), stratified by label, with pixel values normalised to $[0, 1]$. Every rotation condition trains the same two-block CNN (Figure 1). Each block applies Conv(3×3)–BN–ReLU–MaxPool(2×2), expanding channels $1 \to 32 \to 64$ while halving spatial resolution twice ($28^2 \to 14^2 \to 7^2$). The flattened 3,136-dimensional feature vector is projected through FC(128)–ReLU–Dropout(0.5)–FC(10).

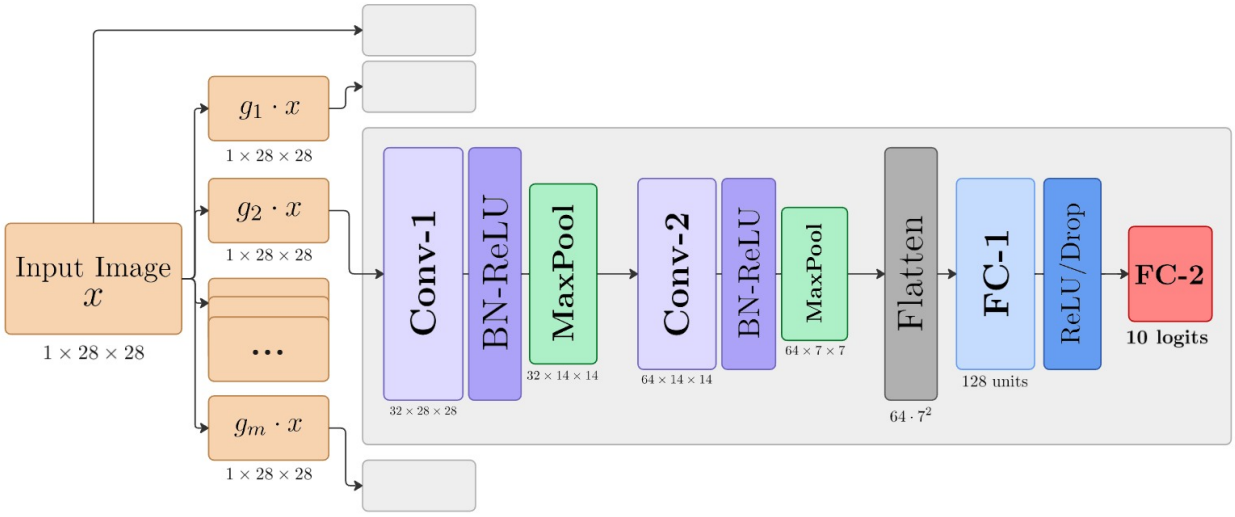


Figure 1: Exemplary flow of a training image $g_j \cdot x$ through the CNN: two convolutional blocks (Conv–BN–ReLU–MaxPool) followed by a two-layer classifier head. Total parameters: 421,834

All conditions are training using a standard Adam optimizer ($\eta = 10^{-3}$, cosine annealing, and $T_{\max} = 15$), batch size 128, 15 epochs, and a fixed random seed. Under Haar augmentation with density m, each mini-batch image x_i is presented alongside m i.i.d. rotated copies $g_1^{(i)} \cdot x_i, \ldots, g_m^{(i)} \cdot x_i$, and the loss is averaged across all mB examples, where B is the mini-batch size:

$$\hat{\mathcal{L}}^{(m)}(\theta) = \frac{1}{mB} \sum_{i=1}^{B} \sum_{j=1}^{m} \ell\Big(f_\theta(g_j^{(i)} \cdot x_i),\, y_i \Big) .$$

All m rotations are batched into a single forward–backward pass, so per-epoch cost scales as $O(m)$.

3.2 Conditions and Metrics

Our baseline condition does not sample rotations ($m = 0$). All other conditions use $m \in \{1, 2, 4, 8, 16\}$ under each of C_4, C_8, and $\mathrm{SO}(2)$ for rotation sampling. For $\mathrm{SO}(2)$, angles are drawn uniformly from $[0, 2\pi)$ and for C_k, we draw rotation angles uniformly from k evenly-spaced rotations.

We measure standard performance metrics, alongside some custom measures:

- **Rotated test accuracy**: each test image is rotated by a fresh $\phi \sim \mathrm{Uniform}[0, 2\pi)$ before prediction, using a fixed seed so conditions are compared on identical angles.
- **Mean rotation robustness**: accuracy averaged over 72 fixed angles in $[0°, 360°)$.
- **Orbit-loss variance**: per-batch variance of per-rotation losses, a proxy for the orbit-gradient variance Σ_g/m (Theorem 2.7).
- **ECE** [9]: calibration quality of the confidence distribution.

4 Experimental Results

We present results addressing questions posed in Section 1.1. Figure use a lighter color shade to present metrics collected during low m sampling density, and vice versa.

4.1 Accuracy Scaling and Group Dependence

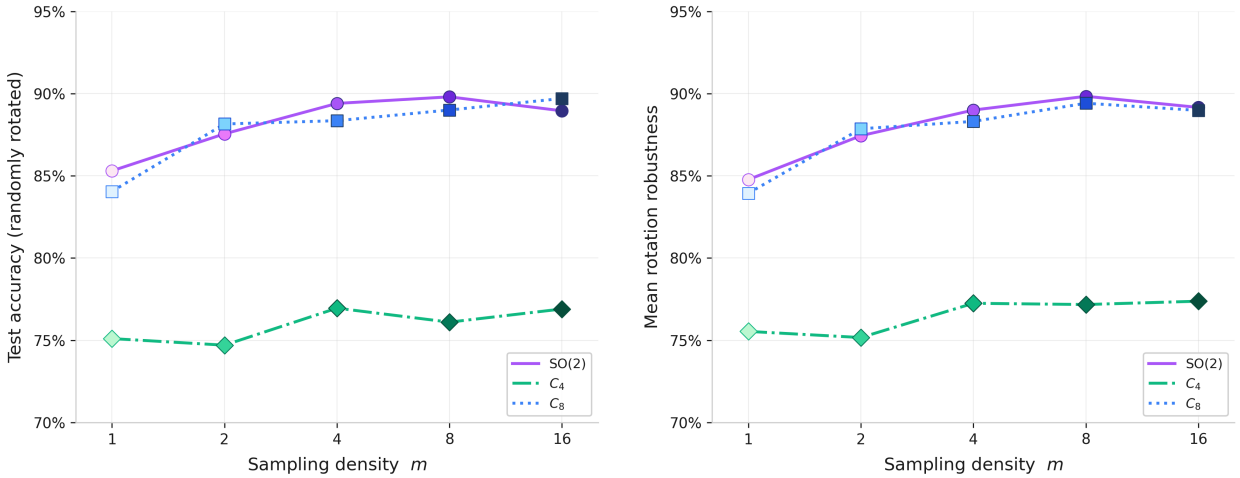


Figure 2: Rotated test accuracy (left) and mean rotation robustness (right) as a function of sampling density m, for $\mathrm{SO}(2)$, C_4, and C_8. Axes omit the baseline (42.4% rotated accuracy) for scale clarity.

For $\mathrm{SO}(2)$, rotated accuracy climbs from 85.3% at $m = 1$ to a peak of 89.8% at $m = 8$, then *dips* to 88.9% at $m = 16$—a hallmark of over-sampling past the useful variance-reduction regime. C_8 follows a similar trajectory, peaking at $m = 16$ (89.7%), consistent with its larger group order requiring more samples to achieve coverage. C_4, by contrast, plateaus near $76{-}77\%$ regardless of m: four discrete orientations cannot generalize to arbitrary $\mathrm{SO}(2)$ test rotations, no matter how densely sampled.

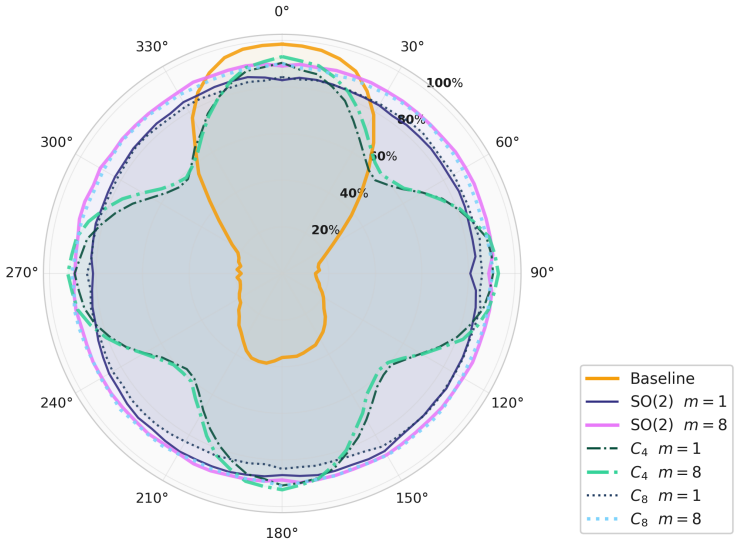


Figure 3: Per-angle accuracy on the full $[0°, 360°)$ rotation range for selected conditions. A perfectly invariant model would trace a filled circle.

The polar diagram (Figure 3) makes the group-coverage argument concrete. The baseline collapses to a narrow spike near $0°$, confirming it has memorized only the canonical orientation. C_4 at $m = 8$ shows four lobes aligned with $0°, 90°, 180°, 270°$—the group's support—with accuracy dropping sharply between them. $\mathrm{SO}(2)$ at $m = 8$ approximates a uniform disc, indicating near-isotropic robustness across all orientations.

4.2 Saturation Threshold and Diminishing Returns

Table 2: Marginal gains in rotated accuracy and mean robustness per doubling of m. A positive entry indicates improvement; negative indicates over-sampling beyond the saturation threshold. Values are percentage points (pp). The final two columns give the additional training seconds required per 1 pp of mean robustness gain relative to $m = 1$ (for $m > 1$); ∞ indicates that the group's performance *declined* at that step.

		Marginal gain (pp) per doubling step				**s / pp rob. vs $m = 1$**	
Group	**Metric**	$1\to2$	$2\to4$	$4\to8$	$8\to16$	$m = 4$	$m = 8$
$\mathrm{SO}(2)$	Acc. (rot.)	$+2.25$	$+1.85$	$+0.40$	-0.85	3 784	7 859
	Mean rob.	$+2.67$	$+1.56$	$+0.84$	-0.68		
C_4	Acc. (rot.)	-0.40	$+2.25$	-0.85	$+0.80$	9 370	24 163
	Mean rob.	-0.37	$+2.08$	-0.07	$+0.20$		
C_8	Acc. (rot.)	$+4.10$	$+0.20$	$+0.65$	$+0.70$	3 657	7 216
	Mean rob.	$+3.92$	$+0.45$	$+1.10$	-0.44		

For $\mathrm{SO}(2)$, the marginal accuracy gain falls from $+2.25$ pp at $m = 1\to2$, to $+1.85$ pp at $m = 2\to4$, to $+0.40$ pp at $m = 4\to8$, and turns negative at $m = 8\to16$—consistent with the $\mathcal{O}(1/\sqrt{m})$ convergence rate of Section 2.2 and indicating that gradient variance is no longer the binding constraint beyond $m = 8$. C_8 saturates later, consistent with $k \ln k \approx 11$ for $k = 8$ (Theorem 2.3), while C_4 exhibits non-monotone behaviour driven by the combinatorial coupon-collector structure of its four discrete elements. Cost efficiency

deteriorates rapidly past saturation: the $\mathrm{SO}(2)$ $m = 8$ peak requires roughly 7,859 s of additional training per 1 pp of robustness gain over $m = 1$, compared to 3,784 s at $m = 4$, identifying $m = 4$ as the most cost-efficient operating point for $\mathrm{SO}(2)$ on MNIST-scale tasks.

4.3 Optimization Effect

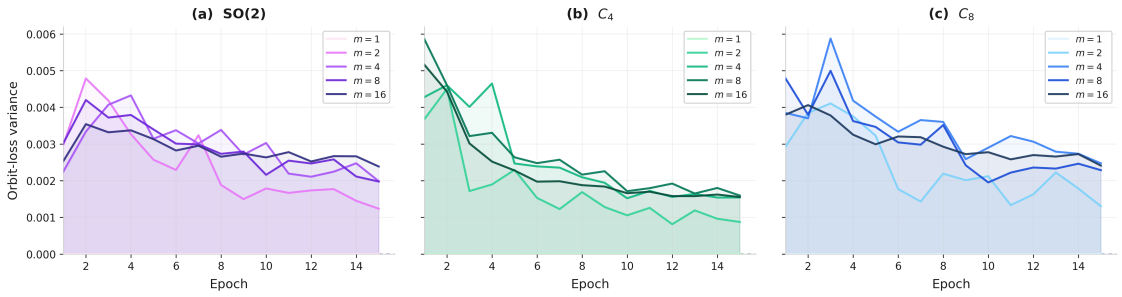


Figure 4: Orbit-loss variance per epoch across all three groups. Dotted horizontal guides mark the theoretical σ_f^2/m level anchored to $m = 1$. All panels share the same y-axis (0–0.006).

Theorem 2.7 predicts that the orbit-gradient variance decays exactly as σ^2/m. Figure 4 provides direct empirical support: for both $\mathrm{SO}(2)$ and C_8, measured orbit-loss variance at epoch 1 halves with each doubling of m, closely tracking the dotted σ_f^2/m guides. C_4 shows similar decay in absolute terms but smaller magnitude, consistent with its narrower orbit (only four elements contribute to the variance). The variance reduction is also visible in the training dynamics: Figure 5 shows that higher m yields markedly smoother loss and accuracy curves.

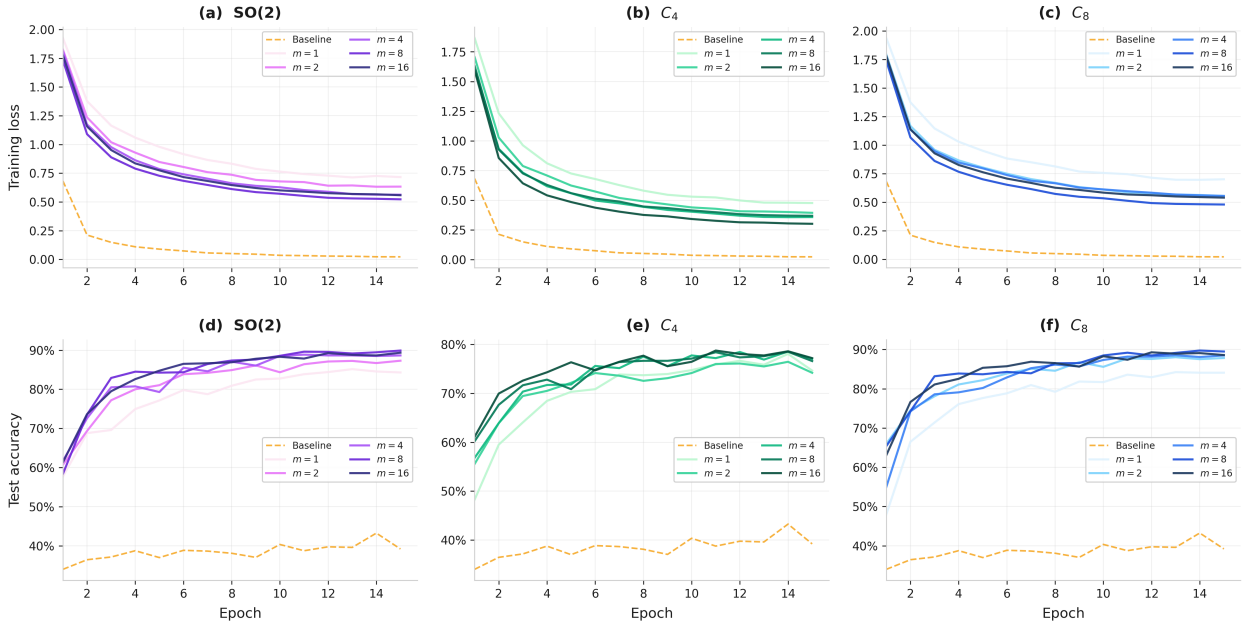


Figure 5: Training loss (top) and rotated test accuracy (bottom) over 15 epochs for all three groups. Higher m (darker shades) produces smoother curves throughout training, reflecting reduced orbit-gradient noise.

4.4 Computational Cost

We confirm that per-epoch cost scales linearly with m across all groups—a direct consequence of the vectorized implementation. We identify $m \in \{4, 8\}$ as the practical operating range for $\mathrm{SO}(2)$ augmentation on MNIST-scale tasks, consistent with the saturation analysis of Section 4.2.

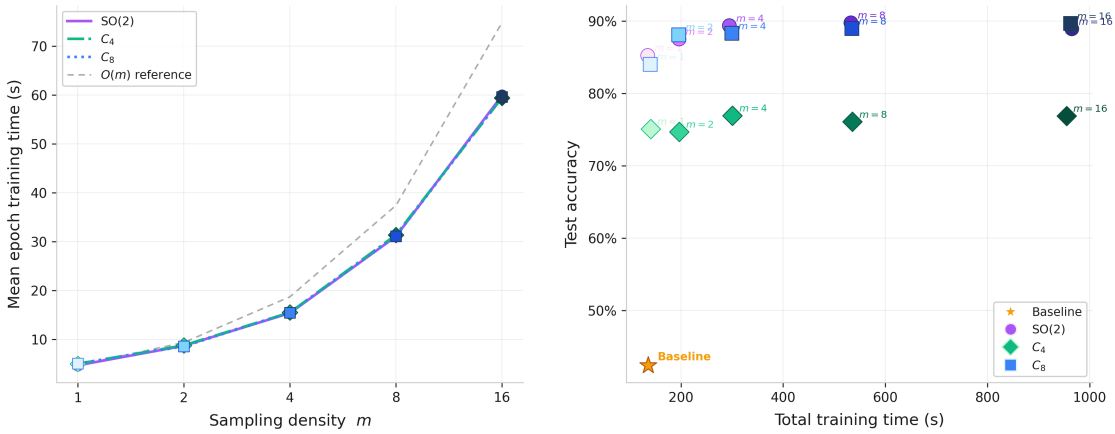


Figure 6: Left: mean epoch training time vs. m, with a linear $O(m)$ reference. Right: rotated test accuracy vs. total training time; each point is one condition, shape encodes group, shade encodes m.

4.5 Summary of Results

Table 3: Primary metrics across all conditions. *Acc. (rot.)* is accuracy on randomly rotated test images (primary); *Acc. (clean)* on non-rotated images (diagnostic); *Mean rob.* averaged over 72 fixed angles.

Group	Cond.	m	Acc. (rot.)	Acc. (clean)	Mean rob.	ECE
None	Baseline	—	0.424	**0.984**	0.400	0.385
$\mathrm{SO}(2)$		1	0.853	0.830	0.848	0.083
		2	0.876	0.868	0.874	0.062
		4	0.894	0.875	0.890	0.055
		8	**0.898**	0.891	**0.898**	**0.039**
		16	0.890	0.882	0.892	0.045
C_4		1	0.751	0.904	0.755	0.033
		2	0.747	0.922	0.752	0.028
		4	0.770	0.926	0.773	0.030
		8	0.761	0.931	0.772	0.028
		16	0.769	0.936	0.774	0.032
C_8		1	0.841	0.843	0.839	0.081
		2	0.882	0.877	0.879	0.057
		4	0.884	0.884	0.883	0.058
		8	0.890	0.901	0.894	0.037
		16	0.897	0.898	0.890	0.057

5 Discussion

The three theoretical predictions of Section 2.6 each find clear empirical support. The $\mathcal{O}(1/\sqrt{m})$ convergence rate (Section 2.2) correctly forecasts the shape of the diminishing-returns curve for $\mathrm{SO}(2)$: gains roughly halve with each doubling of m, and the $\mathcal{O}(1/\sqrt{m})$ guide tracks the empirical curve closely through $m = 8$. The orbit-gradient variance (Theorem 2.7) decays in near exact proportion to $1/m$ across all three groups, confirming that the Monte Carlo gradient estimator behaves as predicted. The coupon-collector bound (Theorem 2.3) correctly orders the saturation thresholds: C_4 saturates earliest ($k \ln k \approx 5.5$), C_8 later ($k \ln k \approx 11$), and $\mathrm{SO}(2)$ never saturates in a combinatorial sense; instead, it follows the smooth variance-reduction regime until $m = 8$ where other factors dominate.

The most striking result is the C_4 ceiling. Regardless of m, C_4 tops out near 76% rotated accuracy—roughly 13 pp below $\mathrm{SO}(2)$ at the same compute budget. This is not a failure of optimization or insufficient data; it is a structural consequence of training on only four orientations when the test distribution is continuous. The result reinforces a principle that architectural equivariant methods make explicit: the invariance built into training must *cover* the invariance demanded at test time. Haar augmentation can approximate this coverage, but only if the sampled group matches the test-time symmetry group.

The decline in $\mathrm{SO}(2)$ accuracy from $m = 8$ to $m = 16$ (-0.85 pp) warrants comment. The $\mathcal{O}(1/\sqrt{m})$ theory predicts diminishing returns but not regression; the dip is therefore not a prediction of the Monte Carlo analysis alone. A plausible explanation is that at large m, the effective batch size grows to $mB = 16 \times 128 = 2{,}048$, which alters the implicit regularization of SGD and may cause underfitting on the $8{,}000$-image training set. This interaction between sampling density and dataset size is not captured by the orbit-variance theory and represents an interesting avenue for future investigation.

For practitioners, the results suggest a simple rule: when training a standard CNN on a task with $\mathrm{SO}(2)$ symmetry, use $m = 4$ as a default. It delivers 89.4% rotated accuracy at $3\times$ the baseline compute—the steepest point on the accuracy-cost curve—and avoids the saturation and potential regression observed at $m \geq 8$. Increasing to $m = 8$ recovers an additional 0.4 pp for $6.5\times$ the baseline cost, which may be worthwhile in accuracy-critical settings but is unlikely to justify the expense at scale. Using a finite group such as C_4 or C_8 is sensible only when the test-time rotation distribution is itself discrete and known in advance.

This study is conducted on MNIST, a low-resolution benchmark where the CNN's capacity is not a limiting factor. It remains to be seen whether the same $\mathcal{O}(1/\sqrt{m})$ scaling and saturation behaviour holds for higher-resolution datasets (e.g. CIFAR-10, ImageNet) and deeper architectures where the orbit variance $\sigma_f^2(x)$ may be substantially different. Additionally, the Hoeffding bound of Theorem 2.4 is known to be loose (Example 2.6): tighter bounds exploiting the actual orbit variance are possible via Bernstein's inequality and would give more informative guidance on the required m in practice.

6 Conclusion

We studied the effect of Haar sampling density m on the rotation robustness of a standard CNN trained with data augmentation. Drawing m i.i.d. group elements per training image and averaging the resulting losses is a Monte Carlo approximation of the true orbit loss, and its quality improves at rate $\mathcal{O}(1/\sqrt{m})$—a rate we derived theoretically and confirmed empirically.

We observed that Haar augmentation with the correct group closes most of the gap between an non-augmented baseline (42.4% rotated accuracy) and a model with near-isotropic robustness (89.8% for $\mathrm{SO}(2)$, $m = 8$), with no architectural changes. Experiments later suggest that the choice of group matters more than the choice of m: C_4 is capped at $\approx 77\%$ accuracy against randomly rotated test images regardless of how densely it is sampled. This is due to the limited range of rotations offered by the group: four discrete orientations cannot

cover a continuous test distribution. Our experiments also confirm that returns diminish rapidly with m, and the practical optimum lies well below the theoretical maximum. A sampling density of $m = 4$ delivers the best accuracy per unit of compute for $\mathrm{SO}(2)$ augmentation on MNIST-scale tasks.

Together, these results build a principled basis for choosing both the augmentation group and the sampling density. To train increased rotation robustness, we should match the rotation group used during augmentation to the test-time symmetry, and stop increasing m once marginal gains fall below a threshold of interest. Future work should examine whether these scaling laws transfer to higher-resolution datasets and deeper architectures, and whether tighter concentration bounds can provide more actionable guidance on m without requiring empirical tuning.

References

[1] Alberto Bietti, Luca Venturi, and Joan Bruna. On the sample complexity of learning under geometric stability. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

[2] Aleksandar Botev, Matthias Bauer, and Soham De. Regularising for invariance to data augmentation improves supervised learning. *arXiv preprint arXiv:2203.03304*, 2022.

[3] Stéphane Boucheron, Gábor Lugosi, and Pascal Massart. *Concentration Inequalities: A Nonasymptotic Theory of Independence*. Oxford University Press, 2013.

[4] Shuxiao Chen, Edgar Dobriban, and Jane H. Lee. A group-theoretic framework for data augmentation. *Journal of Machine Learning Research*, 21(245):1–71, 2020.

[5] Taco S. Cohen and Max Welling. Group equivariant convolutional networks. In *Proceedings of the 33rd International Conference on Machine Learning (ICML)*, 2016.

[6] Taco S. Cohen and Max Welling. Steerable CNNs. In *International Conference on Learning Representations (ICLR)*, 2017.

[7] Bryn Elesedy and Sheheryar Zaidi. Provably strict generalisation benefit for equivariant models. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, 2021.

[8] Gerald B. Folland. *Real Analysis: Modern Techniques and Their Applications*. Wiley, 2nd edition, 1999.

[9] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017.

[10] Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963.

[11] Risi Kondor and Shubhendu Trivedi. On the generalization of equivariance and convolution in neural networks to the action of compact groups. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, 2018.

[12] Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.

[13] Clare Lyle, Mark van der Wilk, Marta Kwiatkowska, Yarin Gal, and Benjamin Bloem-Reddy. On the benefits of invariance in neural networks. *arXiv preprint arXiv:2005.00178*, 2020.

[14] Stéphane Mallat. Group invariant scattering. *Communications on Pure and Applied Mathematics*, 65(10):1331–1398, 2012.

[15] Rajeev Motwani and Prabhakar Raghavan. *Randomized Algorithms*. Cambridge University Press, 1995.

[16] Divya Shanmugam, Davis Blalock, Guha Balakrishnan, and John Guttag. Better aggregation in test-time augmentation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

[17] Bernd Sturmfels. *Algorithms in Invariant Theory*. Springer, 2nd edition, 2008.

[18] Maurice Weiler and Gabriele Cesa. General E(2)-equivariant steerable CNNs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.